

MACHINE LEARNING INTERNSHIP

ASSESSMENT REPORT

Hiring Decision Prediction Using Machine Learning

Submitted To	Kravix Tech Recruitment Team
Candidate	Rithvik
Role Applied	Machine Learning Intern
Dataset	Predicting Hiring Decisions in Recruitment Data
Dataset Source	Kaggle — rabieelkharoua
Models Used	Logistic Regression, Random Forest
Best Accuracy	94.00% (Random Forest)
Date	May 30, 2026

1. Introduction

This report documents the complete end-to-end machine learning project developed for the Kravix Tech ML Internship Assessment. The objective was to build a binary classification model capable of predicting whether a job candidate will be hired, based on various professional and personal attributes. The project demonstrates proficiency in data preprocessing, exploratory data analysis (EDA), model training, evaluation, and comparison — as outlined in the assessment requirements.

2. Problem Statement

Given a dataset of recruitment records containing candidate attributes such as age, education level, years of experience, interview scores, and skill scores, the task is to predict the binary outcome: whether the candidate will be hired (1) or not (0). This is a supervised binary classification problem.

Target Variable

- HiringDecision** — 0 = Not Hired, 1 = Hired

3. Dataset Description

Source

Dataset: *Predicting Hiring Decisions in Recruitment Data*

Platform: Kaggle

Author: rabieelkharoua

URL: kaggle.com/datasets/rabieelkharoua/predicting-hiring-decisions-in-recruitment-data

Dataset Overview

Property	Value
Total Rows	1,500
Total Columns	11 (10 features + 1 target)
Missing Values	0 (completely clean)
Target Column	HiringDecision (0 or 1)
Data Types	int64 (10 columns), float64 (1 column)

Feature Descriptions

Feature	Type	Description
Age	int64	Age of the candidate
Gender	int64	Gender (encoded numerically)
EducationLevel	int64	Highest education level attained
ExperienceYears	int64	Total years of work experience
PreviousCompanies	int64	Number of previous employers
DistanceFromCompany	float64	Commute distance in km
InterviewScore	int64	Score obtained in interview (0-100)

SkillScore	int64	Technical skill assessment score (0-100)
PersonalityScore	int64	Personality evaluation score (0-100)
RecruitmentStrategy	int64	Recruitment channel used
HiringDecision	int64	TARGET: 0 = Not Hired, 1 = Hired

4. Exploratory Data Analysis (EDA)

EDA was conducted to understand the distribution of features, identify patterns, and uncover relationships between variables and the target. The following key insights were discovered:

4.1 Class Balance

The target variable showed a reasonably balanced distribution across the 1,500 records, ensuring no significant bias towards either class during model training.

4.2 Key Feature Insights

- **ExperienceYears:** Candidates with more years of experience had a significantly higher probability of being hired. This was the most visually distinct feature in box plots.
- **InterviewScore:** Hired candidates consistently scored higher in interviews. Strong positive correlation with HiringDecision.
- **SkillScore:** Technical skill scores were notably higher for hired candidates, making this a strong predictive feature.
- **PersonalityScore:** Showed moderate correlation with hiring outcome.
- **Age:** KDE plots revealed that hired candidates tended to cluster in the mid-career age range.
- **EducationLevel:** Higher education levels showed marginally better hiring rates.
- **DistanceFromCompany:** Candidates closer to the company had a slightly higher hire rate.

4.3 Correlation Analysis

A correlation heatmap was generated to identify multicollinearity and feature-target relationships. InterviewScore, SkillScore, and ExperienceYears showed the strongest positive correlations with HiringDecision. No significant multicollinearity was detected among the input features, making all 10 features suitable for model training.

5. Data Preprocessing

5.1 Missing Values

The dataset contained zero missing values across all 11 columns. No imputation or row removal was required.

5.2 Train-Test Split

The dataset was split into training (80%) and test (20%) sets using stratified sampling to maintain class proportions.

- Training set: 1,200 records
- Test set: 300 records
- random_state=42 for reproducibility
- stratify=y to preserve class balance

5.3 Feature Scaling

StandardScaler was applied to normalize all features to zero mean and unit variance. This is critical for Logistic Regression's gradient-based optimization. The scaler was fit on training data only and applied

(transform only) to test data to prevent data leakage.

6. Machine Learning Models

6.1 Model 1 — Logistic Regression

Logistic Regression is a linear baseline model well-suited for binary classification. It models the log-odds of the outcome as a linear combination of input features.

- Solver: lbfgs
- Max iterations: 1000
- Random state: 42
- Regularization: L2 (default)

6.2 Model 2 — Random Forest Classifier

Random Forest is an ensemble method that builds multiple decision trees and aggregates their predictions. It handles non-linear relationships and feature interactions effectively.

- Number of estimators: 100 trees
- Random state: 42
- Bootstrap sampling: enabled
- Feature importance extraction: enabled

7. Evaluation Metrics & Results

7.1 Model Comparison

Metric	Logistic Regression	Random Forest
Accuracy	85.67%	94.00%
ROC-AUC Score	0.8911	0.9287
Best Model		YES

7.2 Interpretation

- **Random Forest outperformed Logistic Regression** on both accuracy (+8.33%) and ROC-AUC (+0.0376), confirming that non-linear decision boundaries exist in this dataset.
- **ROC-AUC of 0.9287** indicates the Random Forest model has excellent discriminatory power — it correctly ranks hired vs. not-hired candidates 92.87% of the time.
- **Logistic Regression** still achieved 85.67% accuracy, validating that linear patterns exist in the data and it serves as a strong baseline.
- **Confusion matrices** were generated for both models to visualize true positives, false positives, true negatives, and false negatives.
- **ROC curves** were plotted for both models on the same axes for direct visual comparison.

7.3 Feature Importance (Random Forest)

Random Forest provides built-in feature importance scores based on mean decrease in impurity. The top predictive features were:

Rank	Feature	Importance
------	---------	------------

1	InterviewScore	Highest
2	SkillScore	High
3	ExperienceYears	High
4	Age	Moderate
5	PersonalityScore	Moderate
6	DistanceFromCompany	Low-Moderate
7	EducationLevel	Low
8	PreviousCompanies	Low
9	RecruitmentStrategy	Low
10	Gender	Lowest

8. Project Structure

The GitHub repository is organized as follows:

```
kravix-ml-assessment/  
  kravix_assessment.ipynb    # Main Jupyter Notebook  
  recruitment_data.csv       # Dataset  
  README.md                  # Project documentation  
  requirements.txt           # Python dependencies  
  Kravix_ML_Report.pdf       # This report
```

9. Code Quality & Practices

- All code is organized into clearly labelled Jupyter Notebook cells with markdown headers.
- Each code cell includes inline comments explaining the logic.
- Functions are kept modular — data loading, EDA, preprocessing, training, and evaluation are separate blocks.
- No data leakage — StandardScaler fit only on training data.
- Stratified train-test split used to preserve class balance.
- Warnings suppressed for clean notebook output.
- All plots saved as high-DPI PNG files (150 dpi).
- requirements.txt lists all dependencies for reproducibility.

10. Conclusion

This project successfully demonstrates a complete machine learning pipeline for predicting hiring decisions. Two classification models were trained and evaluated — Logistic Regression as a linear baseline and Random Forest as an ensemble model.

The Random Forest Classifier emerged as the superior model with **94.00% accuracy** and a **ROC-AUC of 0.9287**, significantly outperforming Logistic Regression. Feature importance analysis confirmed that InterviewScore, SkillScore, and ExperienceYears are the strongest predictors of hiring outcomes — findings that align with real-world recruitment logic.

The project adheres to all assessment criteria: clean code, proper documentation, EDA with visualizations, multiple model comparison, standard evaluation metrics, and professional project structure ready for GitHub submission.